Compiler Design Lab - Assessment 8

Code Optimisation
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Source Code

```
#include <errno.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

static int number(const char *text, long long *value) {
  char *end;
  errno = 0;
  *value = strtoll(text, &end, 10);
  return !errno && end != text && *end == '\0';
}

int main(void) {
  const char *lines[] = {
      "t1 = 4 + 6", "t2 = x + 0", "t3 = 1 * y",
      "t4 = 8 / 2", "t5 = 5 / 0", "t6 = a * b"
  };
  char line[256], dest[64], left[64], right[64], op, extra;
  puts("Original code:");
  for (size_t i = 0; i < sizeof(lines) / sizeof(lines[0]); i++)
    puts(lines[i]);
  puts("\nOptimized code:");
  for (size_t i = 0; i < sizeof(lines) / sizeof(lines[0]); i++) {
    snprintf(line, sizeof(line), "%s\n", lines[i]);
    long long a, b, result;
    if (sscanf(line, " %63s = %63s %c %63s %c", dest, left, &op,
               right, &extra) != 4) {
      fputs(line, stdout);
      continue;
    }

    if (number(left, &a) && number(right, &b)) {
      int valid = 1;
      switch (op) {
      case '+': valid = !__builtin_add_overflow(a, b, &result); break;
      case '-': valid = !__builtin_sub_overflow(a, b, &result); break;
      case '*': valid = !__builtin_mul_overflow(a, b, &result); break;
      case '/':
        valid = b != 0 && !(a == (-9223372036854775807LL - 1) && b == -1);
        if (valid) result = a / b;
        break;
      default: valid = 0;
      }
      if (valid) {
        printf("%s = %lld\n", dest, result);
        continue;
      }
    }

    const char *replacement = NULL;
    if (op == '+' && !strcmp(right, "0")) replacement = left;
    if (op == '+' && !strcmp(left, "0")) replacement = right;
    if (op == '-' && !strcmp(right, "0")) replacement = left;
    if (op == '*' && !strcmp(right, "1")) replacement = left;
    if (op == '*' && !strcmp(left, "1")) replacement = right;
    if (op == '/' && !strcmp(right, "1")) replacement = left;
    if (replacement) printf("%s = %s\n", dest, replacement);
    else printf("%s = %s %c %s\n", dest, left, op, right);
  }
  return 0;
}
```

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Output

```
$ gcc main.c
$ ./a.out
Original code:
t1 = 4 + 6
t2 = x + 0
t3 = 1 * y
t4 = 8 / 2
t5 = 5 / 0
t6 = a * b

Optimized code:
t1 = 10
t2 = x
t3 = y
t4 = 4
t5 = 5 / 0
t6 = a * b
sh-5.3$
```